EXPERIMENT 22

Pronominal Reference Resolution within a Text

AIM

To implement an algorithm for resolving pronominal references (anaphora resolution) within a text to their respective antecedent noun entities.

PROCEDURE

1. Parse the discourse into sentences and extract named entities, nouns, and gender/number traits.
2. Identify candidate third-person pronouns ('he', 'she', 'it', 'they').
3. Apply recency, gender agreement, and syntactic distance heuristics to resolve antecedents.
4. Replace pronouns with resolved antecedents to produce an explicit co-reference mapped text.
5. Display resolved pairs and the expanded text.

PROGRAM

```
class SimpleReferenceResolver:
    def __init__(self):
        self.entity_db = {
            "Alice": {"type": "person", "gender": "female", "number": "sg"},
            "Bob": {"type": "person", "gender": "male", "number": "sg"},
            "The company": {"type": "org", "gender": "neutral", "number": "sg"},
            "The researchers": {"type": "person", "gender": "neutral", "number":
"pl"}
        }

    def resolve(self, discourse):
        sentences = discourse.split('.')
        recent_entities = []
        resolutions = []

        words = discourse.split()
        for i, word in enumerate(words):
            clean_word = word.strip(',.')
            for ent in self.entity_db:
                if ent.lower() == clean_word.lower() or ent.split()[-1].lower() ==
clean_word.lower():
                    recent_entities.append(ent)

            lower_w = clean_word.lower()
            resolved = None
            if lower_w in ['she', 'her']:
                resolved = [e for e in reversed(recent_entities) if
self.entity_db[e]['gender'] == 'female']
            elif lower_w in ['he', 'him', 'his']:
                resolved = [e for e in reversed(recent_entities) if
```

```
self.entity_db[e]['gender'] == 'male']
            elif lower_w in ['it', 'its']:
                resolved = [e for e in reversed(recent_entities) if
self.entity_db[e]['type'] == 'org']
            elif lower_w in ['they', 'them']:
                resolved = [e for e in reversed(recent_entities) if
self.entity_db[e]['number'] == 'pl']

            if resolved:
                resolutions.append((word, resolved[0]))

        return resolutions

discourse = """Alice joined Bob at the conference. She presented her research paper.
Bob commended the presentation. He praised her work. The company funded the travel
because it supports innovation."""

resolver = SimpleReferenceResolver()
resolved_pairs = resolver.resolve(discourse)

print("Original Discourse:\n" + discourse + "\n")
print(f"{'Pronoun (Anaphor)':<18} | {'Resolved Antecedent Referent'}")
print("-" * 50)
for pronoun, antecedent in resolved_pairs:
    print(f"{pronoun:<18} -> {antecedent}")
```

OUTPUT

```
> python3 experiment_22.py
--------------------------------------------------------------------------------
Original Discourse:
Alice joined Bob at the conference. She presented her research paper. Bob commended the
presentation. He praised her work. The company funded the travel because it supports
innovation.

Pronoun (Anaphor)  | Resolved Antecedent Referent
--------------------------------------------------
She                -> Alice
her                -> Alice
He                 -> Bob
her                -> Alice
it                 -> The company
[Process completed with exit code 0]
```

RESULT

Thus, pronominal reference resolution within text was successfully implemented using recency and gender-number constraint matching.